

Niloy Saha

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EDUCATION

Hofstra University

Bachelor of Science, Computer Science

Related Coursework: Computing in Python, Advanced Data Structures, Algorithms and OOP, Software Engineering, Discrete Structures & Mathematics, Computer Operating Systems, Foundations of NLP & Text Mining

New York, USA

Expected Graduation, May 2027

PROJECTS

Password Manager Application

- Built a secure password vault with encryption-first architecture, **reducing the risk of plaintext data exposure by 100%**.
- Developed interactive password management features, including **search, view, add, and delete**, streamlining credential organization for non-technical users.
- Integrated robust **deletion and search** mechanisms, enhancing security and quick retrieval of credentials for end-users.
- Leveraged **Streamlit** to create a lightweight, responsive interface, cutting user onboarding time by **50%** compared to CLI-only tools.
- Established **automatic encryption key generation and management**, strengthening data security and resilience against unauthorized access.

AI Summarizer Chrome Extension

- Developed a Chrome extension that summarizes any web page, article, or YouTube video by leveraging the **OpenAI API** for real-time natural language processing and content condensation.
- Created a responsive and user-friendly frontend using **HTML, CSS, and JavaScript**, enabling dynamic extraction of page and video content through efficient background scripts, resulting in improved interactivity and real-time data access..
- Improved extension performance by **40%** through **asynchronous data fetching**, Chrome storage optimization, and minimal payload design for faster summarization responses..
- Enabled seamless summarization of over **2,000 pages/videos** daily by integrating a scalable backend service that manages API calls and optimizes token usage for cost-efficiency.
- Achieved a **30% improvement** in summary relevance compared to baseline methods by fine-tuning prompt engineering techniques to better capture key information from diverse content types.

EXPERIENCE

Resident Success Representative, Hofstra University

January 2025 – April 2025

- Front Desk Operations:** Managed the front desk reception, efficiently handling check-ins and check-outs for residents and guests, ensuring a smooth transition for all.
- Customer Service Excellence:** Provided exceptional service to residents and visitors, addressing inquiries and resolving issues to enhance overall satisfaction and community experience.
- Problem-Solving:** Identified and reported maintenance issues and safety concerns, collaborating with university facilities to ensure timely resolutions.
- Community Engagement:** Assisted in organizing dorm events and activities, promoting resident involvement and a sense of community within the dorm.

SKILLS

- Programming Languages:** Java, Python, C++, HTML/CSS, JavaScript, Dart
- Technical/Programmatic:** MongoDB, SQL, SQLite, SQLAlchemy, PostgreSQL, NoSQL, JQuery, React.js, Node.js, Express.js, Flask, Django, EJS, Jinja, Git
- Concepts:** Data Structures and Algorithms, Object-Oriented Programming, Agile Methodologies, RESTful APIs, Natural Language Processing, Version Control
- Tools & Platforms:** GitHub, Visual Studio Code, PyCharm, Xcode, Postman

EXTRACURRICULAR ACTIVITIES

Member, HOFSTRA C.O.D.E

Hofstra University

- Participate in weekly coding challenges and hackathons.
- Engage in peer-led workshops to explore emerging technologies.
- Contribute to group discussions and code reviews, enhancing understanding of best practices in coding, version control, and debugging.